

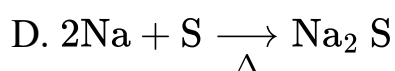
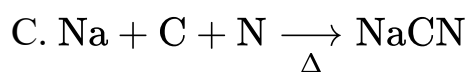
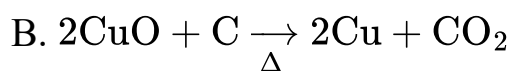
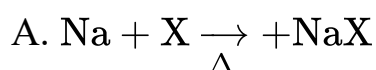
S-block Elements (Alkali & Alkaline Earth Metals)

Question1

Which one of the following reactions does NOT belong to "Lassaigne's test"?

NEET 2025

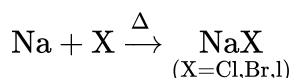
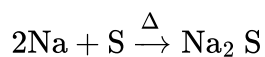
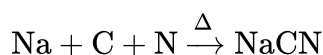
Options:



Answer: B

Solution:

Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by "Lassaigne's test".



Question2

Which one of the following statements is correct?

[NEET 2023]

Options:

A.

All enzymes that utilise ATP in phosphate transfer require Ca as the cofactor

B.

The bone in human body is an inert and unchanging substance

C.

Mg plays roles in neuromuscular function and interneuronal transmission

D.

The daily requirement of Mg and Ca in the human body is estimated to be 0.2 – 0.3g

Answer: D

Solution:

Solution:

All enzymes that utilize ATP in phosphate transfer require Mg as the co-factor.

Bone in human body is not an inert and unchanging substance but is continuously being solubilised and redeposited.

Ca plays important role in neuromuscular function, interneuronal transmission, cell membrane integrity and blood coagulation.

The daily requirement of Mg and Ca in the human body is estimated to be 200 – 300mg (0.2 – 0.3g)

Question3

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which is paramagnetic.

Reason R : The deep blue solution is due to the formation of amide.

In the light of the above statements, choose the correct answer from the options given below :

[NEET 2023]

Options:

A.

Both A and R are true but R is NOT the correct explanation of A

B.

A is true but R is false

C.

A is false but R is true

D.

Both A and R are true and R is the correct explanation of A

Answer: B

Solution:

Solution

On dissolving alkali metal (sodium) in liquid ammonia, a deep blue solution is developed due to ammoniated electron which absorbs energy in visible region of light and imparts blue colour. Due to unpaired electron, solution is paramagnetic.

So, assertion statement is correct but reason is incorrect.

Question4

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) :

Lithium and beryllium unlike their other respective group members form compounds with pronounced ionic character.

Reason (R) :

Lithium and Magnesium have similar properties due to diagonal relationship.

In the light of the above statements, choose the correct answer from the options given below :

[NEET 2023 mpr]

Options:

A.

(A) is true but (R) is false.

B.

(A) is false but (R) is true.

C.

Both (A) and (R) are true and (R) is the correct explanation of (A).

D.

Both (A) and (R) are true but (R) is not the correct explanation of (A).

Answer: B

Solution:

Li, Be forms predominately covalent compounds.

Question5

Identify the incorrect statement from the following
[NEET-2022]

Options:

A. Alkali metals react with water to form their hydroxides.

B. The oxidation number of K in K O_2 is +4.

C. Ionisation enthalpy of alkali metals decreases from top to bottom in the group.

D. Lithium is the strongest reducing agent among the alkali metals.

Answer: B

Solution:

Solution:

Alkali metals show only '+1' oxidation state in all of their compounds.

$K O_2$ is a super-oxide in which O_2^- is anion and K^+ is cation oxidation state of K is +1.

Question6

Match List-I with List-II

List-I	List-II
(a) Li	(i) absorbent for carbon dioxide
(b) Na	(ii) electrochemical cells
(c) KOH	(iii) coolant in fast breeder reactors
(d) Cs	(iv) photoelectric cell

Choose the correct answer from the options given below :
[NEET-2022]

Options:

A. (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

B. (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

C. (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)

D. (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)

Answer: D

Solution:

Solution

- Cs is used in photoelectric cell due to its low ionisation enthalpy
 - KOH used to adsorb CO_2 and changes into K_2CO_3
 - Liquid sodium metal is used as a coolant in fast breeder nuclear reactor
 - Lithium is used in electrochemical cells
-

Question7

$CaCl_2$ and $Ca(OCl)_2$ are components of :
[NEET Re-2022]

Options:

- A. lime water
- B. gypsum
- C. Portland cement
- D. bleaching powder

Answer: D

Solution:

Solution:

$CaCl_2$ and $Ca(OCl)_2$ are components of bleaching powder.

Question8

If first ionization enthalpies of elements X and Y are 419 kJ mol^{-1} and 590 kJ mol^{-1} , respectively and second ionization enthalpies of X

and Y are 3069 kJ mol^{-1} and 1145 kJ mol^{-1} , respectively.

Then correct statement is :

[NEET Re-2022]

Options:

- A. Both X and Y are alkaline earth metals.
- B. X is an alkali metal and Y is an alkaline earth metal.
- C. X is an alkaline earth metal and Y is an alkali metal.
- D. Both X and Y are alkali metals.

Answer: B

Solution:

Solution:

X is alkali metal as it has large size, its IE is less.

After loss of one electron, it gets inert gas configuration. So its IE_2 is very high.

Y is alkaline earth metal. Its IE is more than alkali metal due to stable ns^2 configuration. But its IE_2 is lower than alkali metal.

Question9

Among the following alkaline earth metal halides, one which is covalent and soluble in organic solvents is :

[NEET 2021]

Options:

- A. Calcium chloride
- B. Strontium chloride
- C. Magnesium chloride
- D. Beryllium chloride

Answer: D

Solution:

Solution:

- Except for beryllium chloride all other chloride of alkaline earth metals are ionic in nature.
 - Due to small size of Be, Beryllium chloride is essentially covalent and soluble in organic solvents.
-

Question10

The structures of beryllium chloride in solid state and vapour phase, are :

[NEET 2021]

Options:

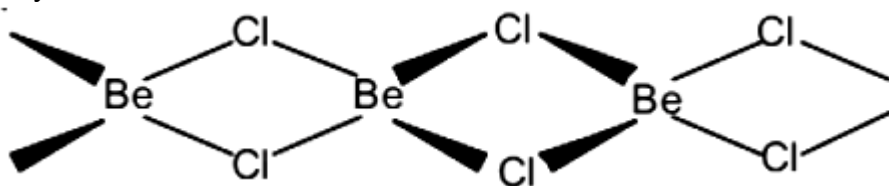
- A. Chain and dimer, respectively
- B. Linear in both
- C. Dimer and Linear, respectively
- D. Chain in both

Answer: A

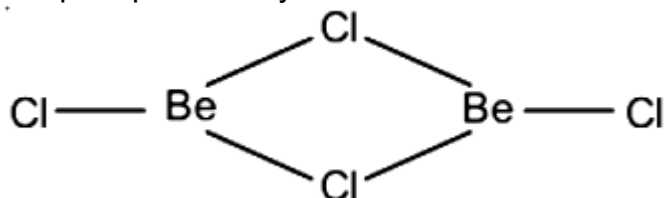
Solution:

Solution:

Beryllium chloride has a chain structure in the solid state as shown below



In vapour phase Beryllium chloride tends to form a chloro-bridged dimer.



Question11

In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it?
[NEET 2021]

©

Options:

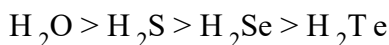
- A. $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: Increasing acidic strength
- B. $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: Increasing pK_a values
- C. $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: Increasing acidic character
- D. $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: Increasing oxidizing power

Answer: B

Solution:

Solution:

Stronger is the acid, lower is the value of pK_a . On moving down the group, bond dissociation enthalpy of hydrides of group 16 elements decreases hence acidity increases and pK_a value decreases. Correct order of pK_a value will be



Question12

The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals.
[2020]

©

Options:

- A. Copper
- B. Calcium
- C. Potassium

D. Iron

Answer: C

Solution:

Solution:

Question13

HCl was passed through a solution of CaCl_2 , MgCl_2 and NaCl . Which of the following compound(s) crystallise(s)? (2020)

©

Options:

A. Only NaCl

B. Only MgCl_2

C. NaCl , MgCl_2 and CaCl_2

D. Both MgCl_2 and CaCl_2

Answer: A

Solution:

Solution:

(a) When HCl is passed through the solution Cl^- ion concentration increases. Hence ionic product becomes more than solubility product. Only NaCl is crystallised due to less solubility than MgCl_2 and CaCl_2

Question14

Enzymes that utilize ATP in phosphate transfer require an alkaline earth metal (M) as the cofactor. M is (NEET 2019)

Options:

- A. Sr
- B. Be
- C. Mg
- D. Ca

Answer: C

Solution:

Solution:

All enzymes that utilise ATP in phosphate transfer require magnesium as the cofactor.

Question15

**Which of the following is an amphoteric hydroxide?
(NEET 2019)**

Options:

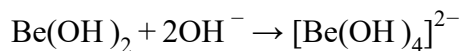
- A. $\text{Be}(\text{OH})_2$
- B. $\text{Sr}(\text{OH})_2$
- C. $\text{Ca}(\text{OH})_2$
- D. $\text{Mg}(\text{OH})_2$

Answer: A

Solution:

Solution:

Be(OH)_2 is amphoteric in nature as it reacts with acid and alkali both.

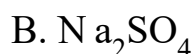


Question16

**Crude sodium chloride obtained by crystallisation of brine solution does not contain
(Odisha NEET 2019)**

©

Options:



Answer: A

Solution:

Solution:

Crude sodium chloride, generally obtained by crystallisation of brine solution contains sodium sulphate (Na_2SO_4), calcium sulphate (CaSO_4), calcium chloride (CaCl_2) and magnesium chloride (MgCl_2) as impurities. Crude sodium chloride does not contain MgSO_4

Question17

**Which of the alkali metal chloride (MCl) forms its dihydrate salt ($\text{MCl} \cdot 2\text{H}_2\text{O}$) easily ?
(Odisha NEET 2019)**

©

Options:

A. LiCl

B. CsCl

C. RbCl

D. KCl

Answer: A

Solution:

Solution:

LiCl is deliquescent and crystallises from aqueous solution as hydrates, $\text{LiCl} \cdot 2\text{H}_2\text{O}$.

Question18

Among CaH_2 , BeH_2 , BaH_2 , the order of ionic character is (NEET 2018)

©

Options:

A. $\text{BeH}_2 < \text{CaH}_2 < \text{BaH}_2$

B. $\text{CaH}_2 < \text{BeH}_2 < \text{BaH}_2$

C. $\text{BeH}_2 < \text{BaH}_2 < \text{CaH}_2$

D. $\text{BaH}_2 < \text{BeH}_2 < \text{CaH}_2$

Answer: A

Solution:

Solution:

$\text{BeH}_2 < \text{CaH}_2 < \text{BaH}_2$

On moving down the group, metallic character of metals increases. So, ionic character of metal hydrides increases. Hence, BeH_2 will be least ionic.

Question19

Magnesium reacts with an element (X) to form an ionic compound. If the ground state electronic configuration of (X) is $1s^2 2s^2 2p^3$, the simplest formula for this compound is (NEET 2018)

Options:



Answer: D

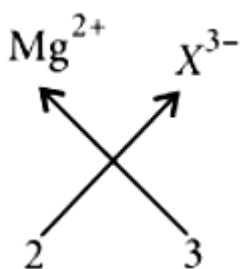
Solution:

Solution:

Electronic configuration of X is

$1s^2, 2s^2 2p^3$. So, valency of X will be 3

Magnesium ion = Mg^{2+}



Formula : Mg_3X_2

Question20

Ionic mobility of which of the following alkali metal ions is lowest when aqueous solution of their salts are put under an electric field? (NEET 2017)

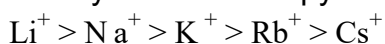
Options:

- A. K
- B. Rb
- C. Li
- D. Na

Answer: C**Solution:**

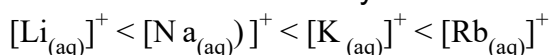
Solution:

The hydration enthalpy of alkali metal ions decreases with increase in ionic sizes i.e.,



Hence, lithium having maximum degree of hydration will be least mobile.

The order of ionic mobility is



Question21

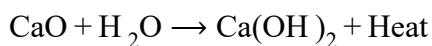
The suspension of slaked lime in water is known as (NEET- II 2016)

Options:

- A. lime water
- B. quick lime
- C. milk of lime
- D. aqueous solution of slaked lime.

Answer: C**Solution:**

Solution:



Quick lime Slaked lime

This process is known as slaking of lime.

The paste of lime in water (i.e.; suspension) is called milk of lime while the filtered and clear solution is known as lime water.

Question22

In context with beryllium, which one of the following statements is incorrect?

(NEET- II 2016)

©

Options:

- A. It is rendered passive by nitric acid.
- B. It forms Be_2C .
- C. Its salts rarely hydrolyse.
- D. Its hydride is electron-deficient and polymeric.

Answer: C

Solution:

Solution:

Beryllium salts are readily hydrolysed.

Question23

Which of the following statements is false?
(NEET- I 2016)

©

Options:

- A. Ca^{2+} ions are not important in maintaining the regular beating of the heart.

B. Mg^{2+} ions are important in the green parts of the plants.

C. Mg^{2+} ions form a complex with ATP.

D. Ca^{2+} ions are important in blood clotting.

Answer: A

Solution:

Solution:

Ca^{2+} ions are required to trigger the contraction of muscles and to maintain the regular beating of the heart.

Question24

The product obtained as a result of a reaction of nitrogen with CaC_2 is
(NEET- I 2016)

©

Options:

A. CaCN_3

B. Ca_2CN

C. $\text{Ca}(\text{CN})_2$

D. CaCN

Answer: C

Solution:

Solution:

Read $\text{Ca}(\text{CN})_2$ as CaCN_2

$\text{CaC}_2 + \text{N}_2 \rightarrow \text{CaCN}_2 + \text{C}$

Nitrolim

Question25

20.0 g of a magnesium carbonate sample decomposes on heating to give carbon dioxide and 8.0 g magnesium oxide What will be the percentage purity of magnesium carbonate in the sample?

(At. wt. of Mg = 24)

(2015)

©

Options:

A. 96

B. 60

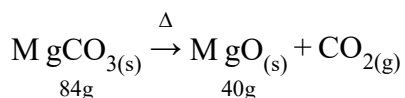
C. 84

D. 75

Answer: C

Solution:

Solution:



84g of MgCO₃ $\equiv$ 40g of MgO

$\therefore$ 20g of MgCO₃ $\equiv \frac{40}{84} \times 20 = 9.52\text{g}$ of MgO

Actual yield = 8g of MgO

$\therefore$ % purity = $\frac{8}{9.52} \times 100 = 84\%$

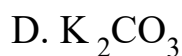
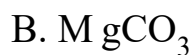
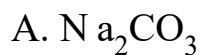
Question26

On heating which of the following releases CO₂ most easily ?

(2015)

©

Options:



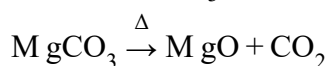
Answer: B

Solution:

Solution:

Stability of carbonates increases down the group with increase in the size of metal ion. Also the alkali metal carbonates are more stable than alkaline earth metal carbonates.

Hence MgCO_3 is least stable and it releases CO_2 most easily

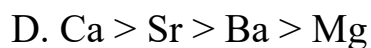
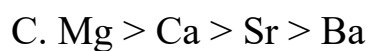
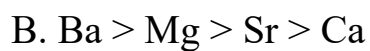
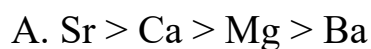


Question27

**Solubility of the alkaline earth metal sulphates in water decreases in the sequence
(2015 Cancelled)**

©

Options:



Answer: C

Solution:

Solution:

Solubility of alkaline earth metal sulphates decreases down the group because hydration energy decreases.

Question28

The function of “Sodium pump” is a biological process operating in each and every cell of all animals. Which of the following biologically important ions is also a constituent of this pump? (2015 Cancelled)

Options:

A. K^+

B. Fe^{2+}

C. Ca^{2+}

D. Mg^{2+}

Answer: A

Solution:

Solution:

Question29

In Castner-Kellner cell for production of sodium hydroxide (Karnataka NEET 2013)

Options:

A. brine is electrolyzed using graphite electrodes

B. molten sodium chloride is electrolysed

C. sodium amalgam is formed at mercury cathode

D. brine is electrolyzed with Pt electrodes.

Answer: C

Solution:

Solution:

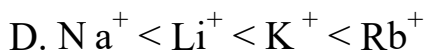
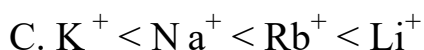
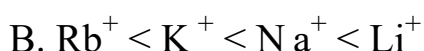
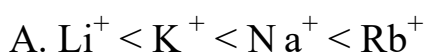
In Castner-Kellner cell, sodium amalgam is formed at mercury cathode.

Question30

**The ease of adsorption of the hydrated alkali metal ions on an ion-exchange resins follows the order:
(2012)**

©

Options:



Answer: B

Solution:

Solution:

In order to explain the behaviour of an atom or ion or molecule, we must have an idea about its size. Atomic radius is the most common way of expressing the size of an atom and similarly the size of an ion is given by the ionic radius.

As we move down in the group 1, the atomic size increases with the increase in the number of shells in the atoms. Similarly the ionic size also shows an increase as we move down the group. So, the order of size of the ions can be given as, Li^+ Smaller is the size of a cation, higher is the charge density on it and thus, higher is the number of water molecules that attach to it in an aqueous solution. As a result, the sizes of the hydrated ions follow the order, Rb^+ Thus, this will also be the order for the ease of adsorption of the ions on ion exchange resins.

Question31

**Which one of the alkali metals, forms only, the normal oxide, M_2O on heating in air ?
(2012)**

Options:

A. Rb

B. K

C. Li

D. Na

Answer: C

Solution:

Solution:

When alkali metals heated in atmosphere of oxygen, the alkali metals ignite and form oxides. On combustion Li forms Li_2O ; sodium gives the peroxide Na_2O_2 and potassium and rubidium give super oxide (MO_2)

Question32

Match List-I with List-II for the compositions of substances and select the correct answer using the code given below.

List - I (Substances)	List - II (Composition)
(A) Plaster of Paris	(i) $CaSO_4 \cdot 2H_2O$
(B) Epsomite	(ii) $CaSO_4 \cdot \frac{1}{2}H_2O$
(C) Kieserite	(iii) $MgSO_4 \cdot 7H_2O$
(D) Gypsum	(iv) $MgSO_4 \cdot H_2O$
	(v) $CaSO_4$

(2011 Mains)

Options:

- A. (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)
 B. (A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)
 C. (A)-(i), (B)-(ii), (C)-(iii), (D)-(v)
 D. (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)

Answer: B

Solution:

Solution:

Plaster of Paris - $CaSO_4 \cdot \frac{1}{2}H_2O$

Epsomite - $MgSO_4 \cdot 7H_2O$

Kieserite - $MgSO_4 \cdot H_2O$

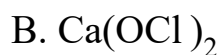
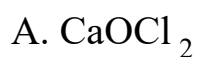
Gypsum - $CaSO_4 \cdot 2H_2O$

Question33

**Which one of the following is present as an active ingredient in bleaching powder for bleaching action?
(2011)**

©

Options:



Answer: B

Solution:

Solution:

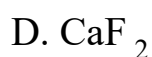
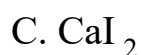
Active ingredient in bleaching powder for bleaching action is $\text{Ca}(\text{OCl})_2$

Question34

**Which of the following compounds has the lowest melting point?
(2011)**

©

Options:



Answer: C

Solution:

Solution:

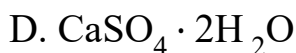
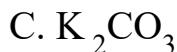
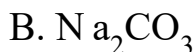
As the covalent character in compound increases and ionic character decreases, melting point of the compound decreases. So, CaI_2 has the highest covalent character and lowest melting point.

Question 35

The compound A on heating gives a colourless gas and a residue that is dissolved in water to obtain B. Excess of CO_2 is bubbled through aqueous solution of B, C is formed which is recovered in the solid form. Solid C on gentle heating gives back A. The compound is (2010 Mains)

©

Options:



Answer: A

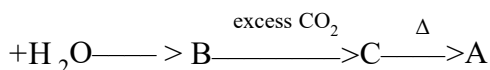
Solution:

Solution:

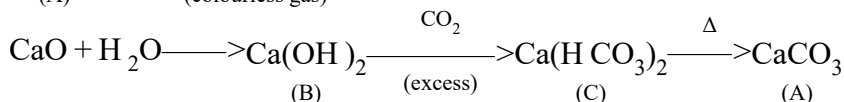
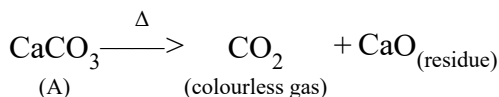
The reactions can be summarised as follows:



Residue



This is possible only when A is CaCO_3 . The reactions are as follows :



Question36

**Which one of the following compounds is a peroxide?
(2010)**

Options:

A. K O_2

B. BaO_2

C. MnO_2

D. NO_2

Answer: B

Solution:

Solution:

BaO_2 has peroxide linkage

Question37

**Property of the alkaline earth metals that increases with their
atomic number
(2010)**

Options:

A. Solubility of their hydroxides in water

B. Solubility of their sulphates in water

C. Ionization energy

D. Electronegativity

Answer: A

Solution:

Solution:

The solubility of an ionic compound depends on two factors :

(a) lattice energy, and (b) hydration energy

In case of alkaline metal hydroxides, the lattice energy decreases as we move down the group.

This decrease is more than the decrease in the hydration energy down the group.

Question38

**Which of the following alkaline earth metal sulphates has hydration enthalpy higher than the lattice enthalpy?
(2010)**

©

Options:

A. CaSO_4

B. BeSO_4

C. BaSO_4

D. SrSO_4

Answer: B

Solution:

Solution:

The hydration enthalpy of BeSO_4 is higher than its lattice energy. Within group 2, the hydration energy decreases down the group while lattice energy is almost the same

Question39

Which of the following oxides is not expected to react with sodium hydroxide?
(2009)

©

Options:

A. CaO

B. SiO₂

C. BeO

D. B₂O₃

Answer: A

Solution:

Solution:

CaO being a basic oxide does not react with NaOH, however SiO₂ (acidic oxide), BeO (amphoteric oxide) and B₂O₃ (acidic oxide) react with NaOH.

Question40

In the case of alkali metals, the covalent character decreases in the order
(2009)

©

Options:

A. MF > MCl > MBr > MI

B. MF > MCl > MI > MBr

C. MI > MBr > MCl > MF

D. MCl > MI > MBr > MF

Answer: C

Solution:

Solution:

Alkali metals are highly electropositive and halogens are electronegative. Thus for the halides of a given alkali metal, the covalent character decreases with increase in electronegativity of halogens.

∴ Order of covalent character of halides is

$\text{MI} > \text{MBr} > \text{MCl} > \text{MF}$

Question 41

The alkali metals form salt-like hydrides by the direct synthesis at elevated temperature. The thermal stability of these hydrides decreases in which of the following orders? (2008)

©

Options:

A. $\text{NaH} > \text{LiH} > \text{KH} > \text{RbH} > \text{CsH}$

B. $\text{LiH} > \text{NaH} > \text{KH} > \text{RbH} > \text{CsH}$

C. $\text{CsH} > \text{RbH} > \text{KH} > \text{NaH} > \text{LiH}$

D. $\text{KH} > \text{NaH} > \text{LiH} > \text{CsH} > \text{RbH}$

Answer: B

Solution:

Solution:

The ionic character of the bonds in hydrides increases from LiH to CsH so thermal stability of these hydrides decreases in the order of

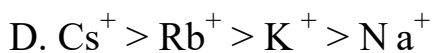
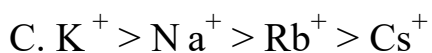
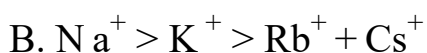
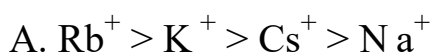
$\text{LiH} > \text{NaH} > \text{KH} > \text{RbH} > \text{CsH}$

Question 42

The sequence of ionic mobility in aqueous solution is (2008)

©

Options:



Answer: D

Solution:

Solution:

Smaller the size of cation, higher will be the hydration and its effective size will increase and hence mobility in aqueous solution will decrease. Hence, the correct sequence of ionic mobility in aqueous solution of the given cations is $\text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{Na}^+$

Question43

Equimolar solutions of the following were prepared in water separately. Which one of the solutions will record the highest pH? (2008)

©

Options:



Answer: D

Solution:

Solution:

Equimolar solutions of the given chlorides when prepared in water forms their respective hydroxides. $\text{Be}(\text{OH})_2$ is amphoteric, but the hydroxides of other alkaline earth metals are basic. The basic strength increases down the group. Hence higher the basic character higher will be the pH

Question44

In which of the following the hydration energy is higher than the lattice energy?
(2007)

Options:

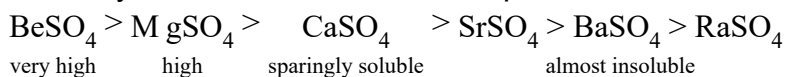


Answer: A

Solution:

Solution:

When hydration energy exceeds lattice energy, the compound becomes soluble in water. The solubility of alkaline earth metal sulphates decreases in the order



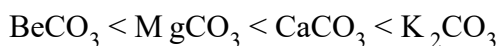
The solubilities of BeSO_4 and MgSO_4 are due to high energy of solvation of smaller Be^{2+} and Mg^{2+} ions

Question45

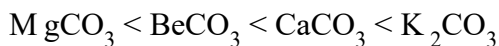
The correct order of increasing thermal stability of K_2CO_3 , MgCO_3 , CaCO_3 and BeCO_3 is
(2007)

Options:

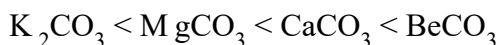
A.



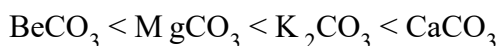
B.



C.



D.

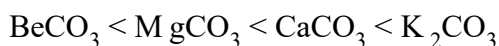
**Answer: A****Solution:**

Solution:

In all cases, for a particular set of e.g. Group 1 or Group 2 compounds, the thermal stability increases down the group as the ionic radius of the cation increases, and its polarising power decreases.

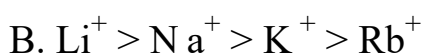
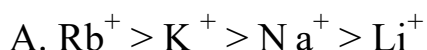
Group 1 compounds tend to be more thermally stable than group 2 compounds because the cation has a smaller charge and a larger ionic radius, and so a lower polarising power, particularly when adjacent metals on the same period are compared.

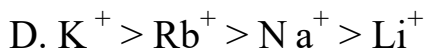
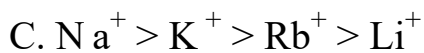
Hence the order of increasing thermal stability of K_2CO_3 , MgCO_3 , CaCO_3 and BeCO_3 is



Question 46

**The correct order of the mobility of the alkali metal ions in aqueous solution is
(2006)**

Options:

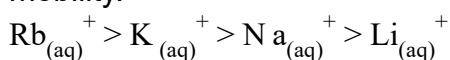


Answer: A

Solution:

Solution:

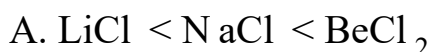
The alkali metal ions exist as hydrated ions $\text{M}^+(\text{H}_2\text{O})_x$ in the aqueous solution. The degree of hydration, however, decreases with the ionic size as we move from Li^+ to Cs^+ . In other words, Li^+ ion is most highly hydrated, e.g. $[\text{Li}(\text{H}_2\text{O})_6]^+$. Since the mobility of ion is inversely proportional to the size of their hydrated ions, therefore, amongst the alkali metal ions, lithium has the lowest ionic mobility.



Question 47

The correct sequence of increasing covalent character is represented by
(2005)

Options:



Answer: C

Solution:

Solution:

Covalent character in a compound is found by Fajan's Rule.

Fajan's Rule :

(i) Greater the size of the cation more will be the ionic character and

(ii) Greater the size of anion more will be its covalent character and charge on the cation is

dominant over the size of the cation.

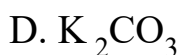
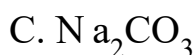
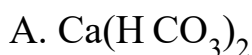
(iii) Greater the charge on the cation more will be its covalent character.

Hence, $\text{BeCl}_2 > \text{LiCl} > \text{NaCl}$ (In Covalent character).

Question48

A solid compound X on heating gives CO_2 gas and a residue. The residue mixed with water forms Y. On passing an excess of CO_2 through Y in water, a clear solution Z is obtained. On boiling Z, compound X is reformed. The compound X is (2004)

Options:

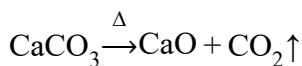


Answer: B

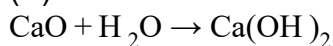
Solution:

Solution:

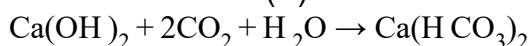
The given compound X must be CaCO_3 . It can be explained by following reactions:



(X)



(Y)



(Z)



(X)

Question49

In which of the following processes, fused sodium hydroxide is electrolysed at a 330°C temperature for extraction of sodium? (2000)

©

Options:

- A. Castner's process
- B. Down's process
- C. Cyanide process
- D. Both (b) and (c).

Answer: A

Solution:

Solution:

In Castner's process, for production of sodium metal, sodium hydroxide (NaOH) is electrolysed at temperature 330°C .

Question50

When a substance (A) reacts with water it produces a combustible gas (B) and a solution of substance (C) in water. When another substance (D) reacts with this solution of (C), it also produces the same gas (B) on warming but (D) can produce gas (B) on reaction with dilute sulphuric acid at room temperature. Substance (A) imparts a deep golden yellow colour to a smokeless flame of Bunsen burner. Then (A), (B), (C) and (D) respectively are (1998)

©

Options:

A. Ca, H₂, Ca(OH)₂, Sn

B. K, H₂, KOH, Al

C. Na, H₂, NaOH, Zn

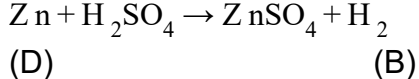
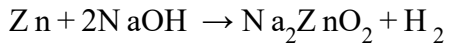
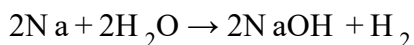
D. CaC₂, C₂H₂, Ca(OH)₂, Fe

Answer: C

Solution:

Solution:

Only 'Na' imparts golden colour to Bunsen flame, therefore, A = Na, B = H₂, C = NaOH, D = Zn



Question 51

**Calcium is obtained by the
(1997)**

Options:

A. electrolysis of solution of calcium chloride in water

B. electrolysis of molten anhydrous calcium chloride

C. roasting of limestone

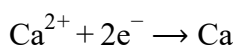
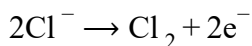
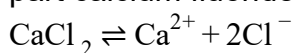
D. reduction of calcium chloride with carbon.

Answer: B

Solution:

Solution:

Calcium is obtained by electrolysis of a fused mass consisting six parts calcium chloride and one part calcium fluoride at about 700°C



Question52

Sodium is made by the electrolysis of a molten mixture of about 40%NaCl and 60%CaCl₂ because (1995)

©

Options:

- A. Ca^{++} can reduce NaCl to Na
- B. Ca^{++} can displace Na from NaCl
- C. CaCl_2 helps in conduction of electricity
- D. this mixture has a lower melting point than NaCl .

Answer: D

Solution:

Solution:

Sodium is obtained by electrolytic reduction of its chloride. Melting point of chloride of sodium is high so in order to lower its melting point, calcium chloride is added to it.

Question53

**The solubility in water of sulphate down the Be group is
 $\text{Be} > \text{Mg} > \text{Ca} > \text{Sr} > \text{Ba}$.**

**This is due to
(1995)**

©

Options:

- A. decreasing lattice energy
- B. high heat of solvation for smaller ions like Be^{2+}
- C. increase in melting points
- D. increasing molecular weight.

Answer: B

Solution:

Solution:

As we move down the group from BeSO_4 to BaSO_4 the enthalpy of hydration of the positive ion becomes smaller due to increase in ionic size. Salts of heavier metal ions are less soluble than those of lighter ions.

Question54

**Identify the correct statement.
(1995)**

©

Options:

- A. Plaster of Paris can be obtained by hydration of gypsum.
- B. Plaster of Paris is obtained by partial oxidation of gypsum.
- C. Gypsum contains a lower percentage of calcium than Plaster of Paris.
- D. Gypsum is obtained by heating Plaster of Paris.

Answer: C

Solution:

Solution:

Gypsum is $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ and Plaster of Paris is $(\text{CaSO}_4)_2 \cdot \text{H}_2\text{O}$. Therefore gypsum contains a lower percentage of calcium than Plaster of Paris.

Question55

**Which of the following is known as fusion mixture?
(1994)**

©

Options:

A. Mixture of $\text{Na}_2\text{CO}_3 + \text{NaHCO}_3$

B. $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

C. Mixture of $\text{K}_2\text{CO}_3 + \text{Na}_2\text{CO}_3$

D. NaHCO_3

Answer: C

Solution:

Solution:

K_2CO_3 and Na_2CO_3 mixture is called as fusion mixture.

Question56

**All the following substances react with water. The pair that gives the same gaseous product is
(1994)**

©

Options:

A. K and KO_2

B. Na and Na_2O_2

C. Ca and CaH_2

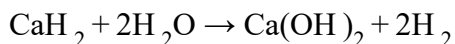
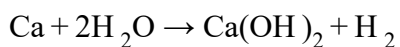
D. Ba and BaO_2 .

Answer: C

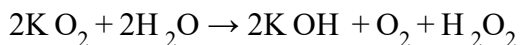
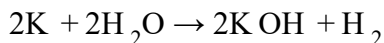
Solution:

Solution:

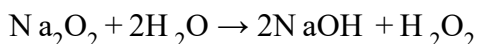
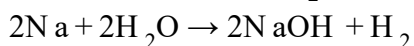
The pair which gives the same gaseous product is Ca and CaH_2 .



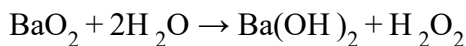
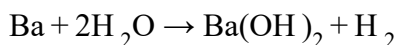
Whereas, K gives H_2 while K_2O_2 gives O_2 and H_2O_2



Similarly, Na gives H_2 , while Na_2O_2 gives H_2O_2 .



Likewise Ba gives H_2 while BaO_2 gives H_2O_2 .



Question 57

Among the following oxides, the one which is most basic is (1994)

©

Options:

A. ZnO

B. MgO

C. Al_2O_3

D. N_2O_5

Answer: B

Solution:

Solution:

Al_2O_3 and ZnO are amphoteric. N_2O_5 is strongly acidic.

Question58

Which of the following metal ions play an important role in muscle contraction?
(1994)

©

Options:

A. K^{+}

B. Na^{+}

C. Mg^{2+}

D. Ca^{2+}

Answer: D

Solution:

Solution:

Calcium is an essential element for the contraction of muscles. In the presence of calcium ions and energy from ATP, actin and myosin (contractile proteins) interact forming actomyosin which causes contraction of muscles.

Question59

Which of the following statement is false?
(1994)

©

Options:

A. Strontium decomposes water readily than beryllium.

B. Barium carbonate melts at a higher temperature than calcium carbonate.

C. Barium hydroxide is more soluble in water than magnesium hydroxide.

D. Beryllium hydroxide is more basic than barium hydroxide.

Answer: D

Solution:

Solution:

Beryllium hydroxide although amphoteric, is however less basic than barium hydroxide.

Question60

**Which one of the following has minimum value of cation/anion ratio?
(1993)**

©

Options:

A. NaCl

B. KCl

C. $MgCl_2$

D. CaF_2

Answer: C

Solution:

Solution:

The order of ionic size for given ions will be $K^+ > Ca^{2+} > Mg^{2+}$ and that of $Cl^- > F^-$ Therefore, Mg^{2+}/Cl^- has minimum value of cation/anion ratio.

Question61

**Which of the following has largest size?
(1993)**

©

Options:

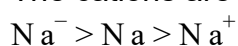
- A. Na
- B. Na^+
- C. Na^-
- D. Can't be predicted.

Answer: C

Solution:

Solution:

The cations are always smaller than the neutral atom and anions are always larger in size



Question62

**Which compound will show the highest lattice energy?
(1993)**

©

Options:

- A. RbF
- B. CsF
- C. NaF
- D. KF

Answer: C

Solution:

Solution:

With the same anion, smaller the size of the cation, higher is the lattice energy. Therefore, NaF will show the highest lattice energy among the given compounds.

Question63

Strongest bond is in between (1993)

©

Options:

- A. CsF
- B. NaCl
- C. both (a) and (b)
- D. none of the above.

Answer: A

Solution:

Solution:

According to Fajan rules, ionic character increases with increase in size of the cation, ($\text{Cs} > \text{Rb} > \text{K} > \text{Na}$) and with decrease in size of the anion ($\text{F} > \text{Cl} > \text{Br} > \text{I}$). Thus, CsF has higher ionic character than NaCl and hence bond in CsF is stronger than in NaCl.

Question 64

Compared with the alkaline earth metals, the alkali metals exhibit (1993)

©

Options:

- A. smaller ionic radii
- B. highest boiling points
- C. greater hardness
- D. lower ionization energies.

Answer: D

Solution:

Solution:

The alkali metals are larger in size and have smaller nuclear charge thus they have lower ionization energy in comparison to alkaline earth metals.

Question65

Electronic configuration of calcium atom may be written as (1992)

©

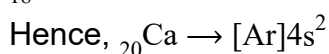
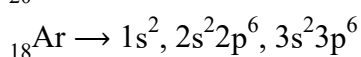
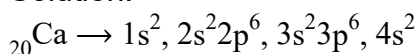
Options:



Answer: B

Solution:

Solution:



Question66

Which one of the following substance is used in the laboratory for fast drying of neutral gases? (1992)

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Options:

A. Phosphorus pentoxide

B. Active charcoal

C. Anhydrous calcium chloride

D. Na_3PO_4

Answer: A

Solution:

Solution:

P_2O_5 absorbs moisture much readily than anhydrous CaCl_2 .

Question67

**Washing soda has formula
(1990)**

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Options:

A. $\text{Na}_2\text{CO}_3 \cdot 7\text{H}_2\text{O}$

B. $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

C. $\text{Na}_2\text{CO}_3 \cdot 3\text{H}_2\text{O}$

D. Na_2CO_3

Answer: B

Solution:

Solution:

$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ is washing soda.

Question68

Which one of the following properties of alkali metals increases in magnitude as the atomic number rises?
(1989)

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Options:

- A. Ionic radius
- B. Melting point
- C. Electronegativity
- D. First ionization energy

Answer: A

Solution:

Solution:

In a group, ionic radius increases with increase in atomic number whereas the m.pt. decreases down in a group due to weakening of metallic bond. Similarly, electronegativity and the ionization energy also decreases down the group.

Question69

Which of the following atoms will have the smallest size?
(1989)

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Options:

- A. Mg
- B. Na
- C. Be
- D. Li

Answer: C

Solution:

Solution:

The atomic size decreases within a period from left to right, therefore $\text{Li} > \text{Be}$ and $\text{Na} > \text{Mg}$. The size increases in a group from top to bottom. Hence, the size of Na is greater than Li. Overall order $\text{Na} > \text{Mg} > \text{Li} > \text{Be}$. Thus, Be has smallest size.

Question 70

Bleaching powder is obtained by the action of chlorine gas and (1988)

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Options:

- A. dilute solution of Ca(OH)_2
- B. concentrated solution of Ca(OH)_2
- C. dry CaO
- D. dry slaked lime.

Answer: D

Solution:

Solution:

Cl_2 gas reacts with dry slaked lime, Ca(OH)_2 to give bleaching powder.

